Matrices & Systems

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01

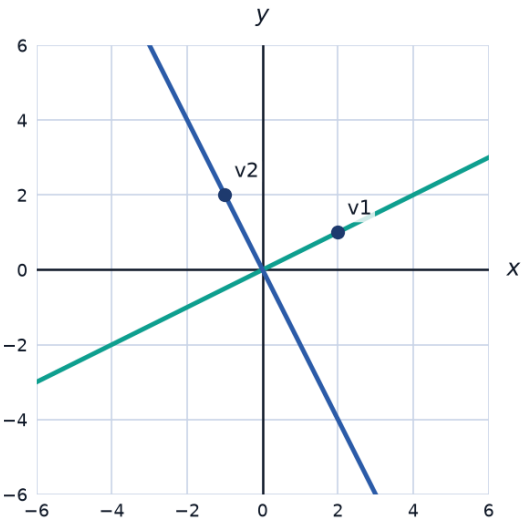
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Reciprocal, quotient, and Pythaogrean identities**

Given the matrices $A = [4, -2; 1, 3]$ and $B = [-1, 5; 2, 0]$, what is the matrix sum $A + B$?

2 **[Foundational] • Target: Cofunction identities**

Given the matrices $A = (3, -2; 1, 4)$ and $B = (-5, 2; 0, -3)$, what is the matrix $2A - B$?



3 **[Foundational] • Target: Angle sum and difference identities**

In the figure, right triangle ABC has a right angle at vertex C, and altitude CD is drawn perpendicular to hypotenuse AB. If $AD = x$, $DB = x + 5$, and $CD = 6$, what is the length of hypotenuse AB?

4 **[Foundational] • Target: Verifying trig identities**

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

5 **[Foundational] • Target: Reciprocal, quotient, and Pythaogrean identities**

In the xy-plane, an angle θ is in standard position. If the terminal side of θ passes through the point $(-5, 12)$, what is the value of $\sin(\theta)$?

6 **[Foundational] • Target: Cofunction identities**

In a right-angled triangle, one of the acute angles measures 30°. If the side opposite this angle has a length of 5 units, what is the length of the hypotenuse?

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7 [Foundational] • Target: Angle sum and difference identities

Given the matrices $A = (5, -1; 2, 4)$ and $B = (k, 2; -2, 5)$, for what value of the constant k is the matrix $C = 2A - B$ not invertible?

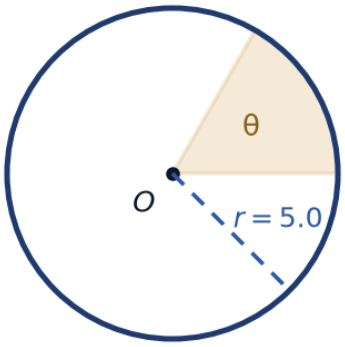
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Verifying trig identities

An arithmetic sequence has a first term of $a_1 = 5$ and a common difference of $d = 3$. What is the value of the 6th term, a_6?

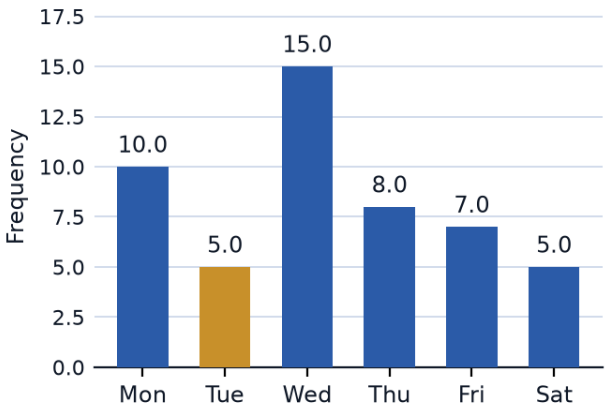
9 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

What are the coordinates of the center and the radius of the circle given by the equation $(x - 4)^2 + (y + 1)^2 = 36$?



10 [Intermediate] • Target: Cofunction identities

A small bakery recorded the number of muffins sold each day for a week. The results are shown in the bar chart below. What is the total number of muffins sold during the week?



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11 [Intermediate] • Target: Angle sum and difference identities

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $(x - 2)$. When $P(x)$ is divided by $(x + 1)$, the remainder is -12. What is the remainder when $P(x)$ is divided by $(x - 3)$?

12 [Intermediate] • Target: Verifying trig identities

An ellipse in the xy-plane is defined by the equation $9x^2 + 16y^2 - 54x + 64y + c = 0$ where c is a constant. If the length of the minor axis of the ellipse is 6, what is the value of c?

13 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

The polynomial $P(x) = x^4 + ax^3 - 7x^2 + bx - 12$ has zeroes at $x = -3$ and $x = 2$, where a and b are real constants. What is the value of $a + b$?

14 [Intermediate] • Target: Cofunction identities

Consider the sequence a_n defined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$ for $n \geq 1$. If the sequence converges to a limit L, find the value of $L^2 - 2L$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A hyperbola is defined by the equation $\frac{(x-h)^2}{16} - \frac{(y-k)^2}{9} = 1$. If the vertices of the hyperbola are located at $(5, 2)$ and $(-1, 2)$, determine the values of h and k.

16 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 6.01 Radicals & Rational Exponents

For all real values of x such that $xeq -1$ and $xeq2$, the equation $\frac{ax+7}{x+1} - \frac{b}{x-2} = \frac{2x^2 - 3x - 20}{x^2 - x - 2}$ is true, where a and b are constants. What is the value of $a + b$?

17 [Challenging] • Target: Reciprocal, quotient, and Pythaogrean identities • Cross-Lesson: 7.01 Sequences & Series

A manufacturing process produces items, and the number of defects per batch of 100 items follows a Poisson distribution with a mean of λ. The cost of inspecting a batch is C_i dollars, and the cost of repairing a defective item is C_r dollars. If a batch has k defects, the total cost for that batch is $C(k) = C_i + k \cdot C_r$. If $\lambda = 3$, $C_i = 50$, and $C_r = 10$, find the expected total cost per batch, given that the expected number of defects in a batch is $E[K] = \lambda$.

18 [Challenging] • Target: Cofunction identities • Cross-Lesson: 8.01 Conic Sections

Consider the equation $\sqrt{2x + k} = x - 1$, where k is a constant. If $x = 5$ is a valid solution to the equation, what is the extraneous solution that arises when solving the equation algebraically by squaring both sides?

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19 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 9.01 Trigonometric Functions

A set of exam scores is normally distributed with a mean of $\mu = 74$ and a standard deviation of $\sigma = 8$. What is the z-score corresponding to an exam score of 58?

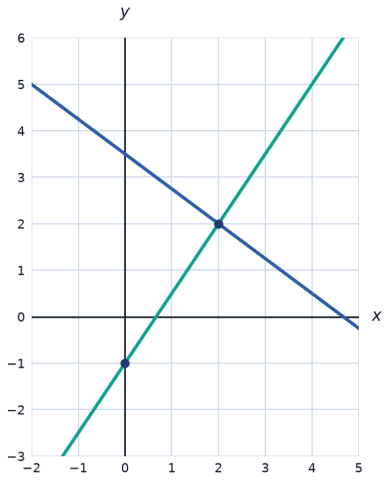
20 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 1.01 Functions & Their Graphs

Let $f(x) = \sqrt{x^2 - 6x + 9}$ and $g(x) = x - 3$. For which values of x is $f(x) = g(x)$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reciprocal, quotient, and Pythaogrean identities

For the model $f(x) = 2x + 5$ in reciprocal, quotient, and pythaogrean identities, evaluate $f(2)$.



a. 9

b. 11

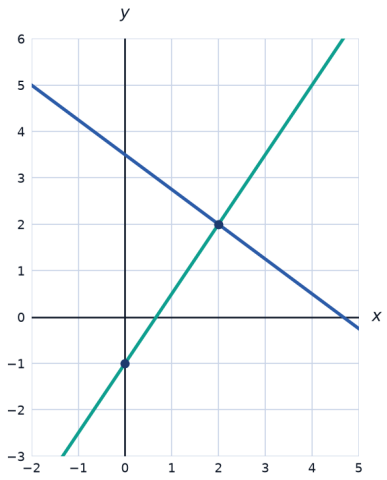
c. 4

d. 13

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22 [Foundational] • Target: Reciprocal, quotient, and Pythaogrean identities

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

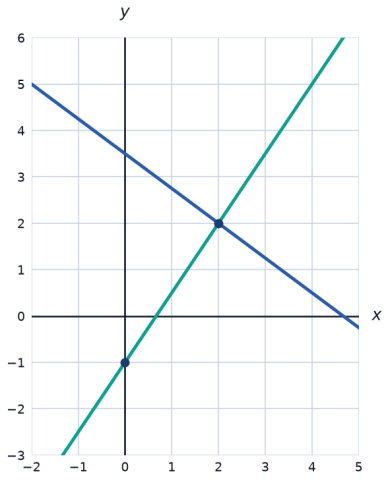
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

23 [Foundational] • Target: Reciprocal, quotient, and Pythaogrean identities

For the model $f(x) = 2x + 5$ in reciprocal, quotient, and pythaogrean identities, evaluate $f(5)$.



a. 15

b. 17

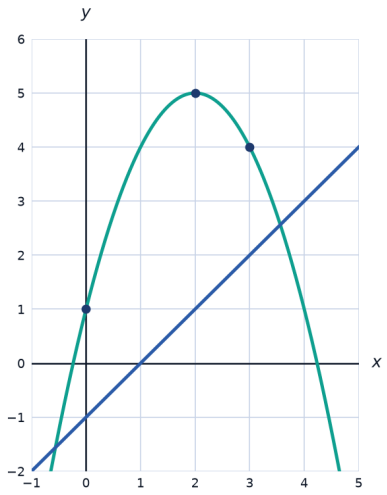
c. 10

d. 19

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24 [Foundational] • Target: Cofunction identities

For the model $f(x) = 3x + 6$ in cofunction identities, evaluate $f(2)$.



a. 12

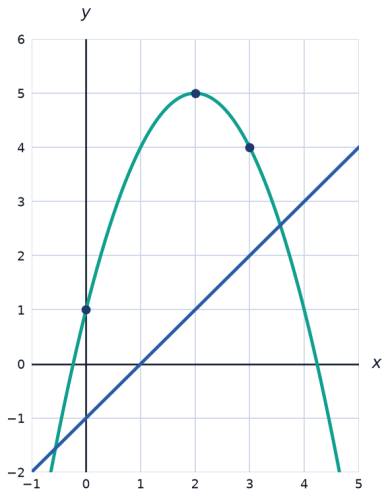
b. 15

c. 6

d. 16

25 [Foundational] • Target: Cofunction identities

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

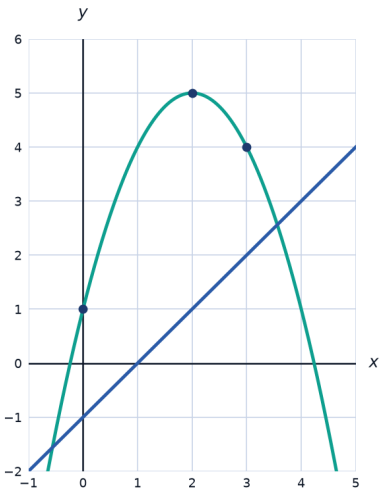
c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

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26 [Foundational] • Target: Cofunction identities

For the model $f(x) = 3x + 6$ in cofunction identities, evaluate $f(5)$.



a. 21

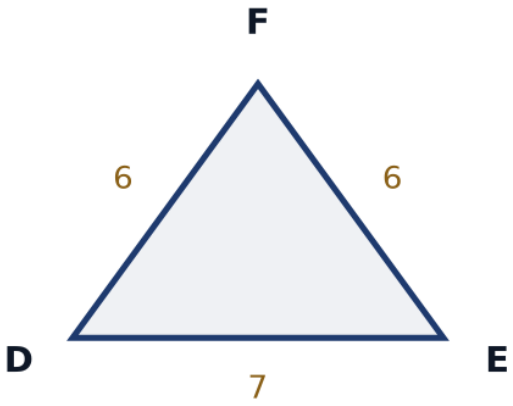
b. 24

c. 15

d. 25

27 [Foundational] • Target: Angle sum and difference identities

In ΔABC studying angle sum and difference identities, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 88°

b. 98°

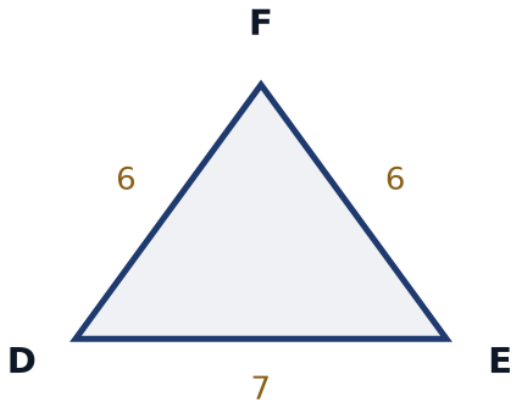
c. 80°

d. 133°

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28 [Foundational] • Target: Angle sum and difference identities

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

b. 18 cm

c. 24 cm

d. 54 cm

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

30 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

A data set recorded for reciprocal, quotient, and pythaogrean identities consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

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31 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

For the model $f(x) = 2x + 5$ in reciprocal, quotient, and pythaogrean identities, evaluate $f(4)$.

a. 13

b. 15

c. 8

d. 17

32 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

33 [Intermediate] • Target: Cofunction identities

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

34 [Intermediate] • Target: Cofunction identities

A data set recorded for cofunction identities consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

35 [Intermediate] • Target: Cofunction identities

For the model $f(x) = 3x + 6$ in cofunction identities, evaluate $f(4)$.

a. 18

b. 21

c. 12

d. 22

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36 [Intermediate] • Target: Cofunction identities

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

37 [Intermediate] • Target: Angle sum and difference identities

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 8.01 Conic Sections

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 14

c. 12

d. 17

39 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 9.01 Trigonometric Functions

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 12

b. 19

c. 17

d. 22

40 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 1.01 Functions & Their Graphs

In ΔABC studying angle sum and difference identities, $m\angle A = 47^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 74°

b. 84°

c. 66°

d. 133°

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41 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 11.01 Probability & Statistics

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

42 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 2.01 Quadratics & Complex Numbers

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 22

b. 29

c. 27

d. 32

43 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 3.01 Polynomial Functions

A data set recorded for verifying trig identities consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

44 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 5x + 8$ in verifying trig identities, evaluate $f(3)$.

a. 23

b. 28

c. 15

d. 27

45 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

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